

ECG Anomaly Detection — Session Report

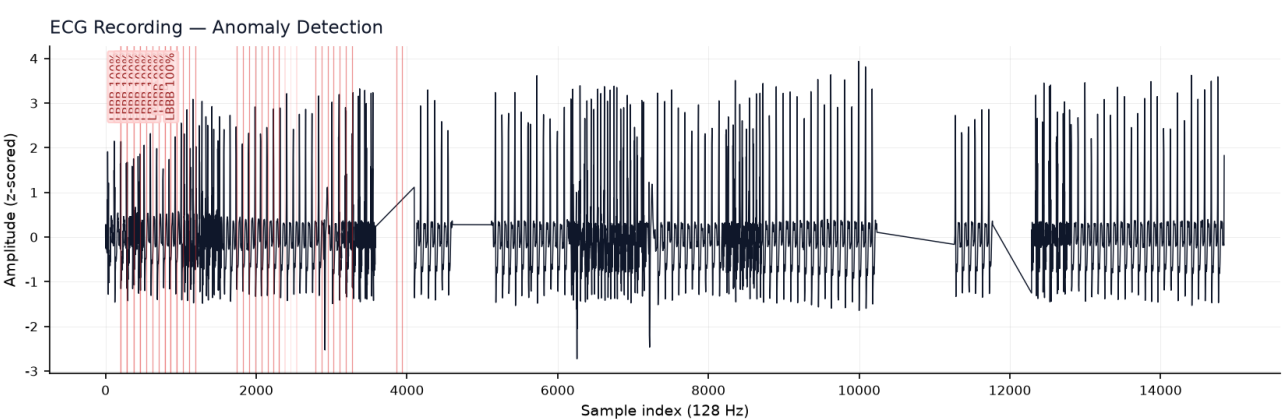
Prepared for: II • Session ID: 584387c3 • Generated: 2026-07-29 06:55:25

Session Information

Patient / User	Sriyantha Sadaruwan
Source	mit-bih-arrhythmia (109)
Started	2026-07-29 06:55:25
Ended	2026-07-29 06:56:13
Duration	43.4 s
Status	completed
Total beats analyzed	234
Anomalous beats	234 (100.00%)
Model	ecg-ae-v2 (v1783189189) • Dilated-U-Net-Autoencoder
Analysis mode	Beat-aligned hierarchical morphology + RR
Abnormal-probability threshold	0.3735

ECG Recording & Detection Results

The following plot shows the recorded ECG signal (z-score normalized). Red markers indicate R-peak-aligned beats whose supervised abnormal probability exceeded the validation-selected threshold.



Main Arrhythmia Finding

Left Bundle Branch Block was the main high-confidence arrhythmia type detected in this session. It appeared 175 time(s) with average confidence 99.9% and maximum confidence 100.0%. Reason: the beat-aligned detector combined morphology, spectrum and RR timing, and its abnormal probability exceeded the saved threshold. The subtype model then assigned this label with at least 50% confidence. Lower-confidence labels are excluded to avoid over-interpreting uncertain model output.

Example: time 2026-07-29 06:56:04, abnormal probability 0.9120, threshold 0.3735, classifier confidence 100.0%.

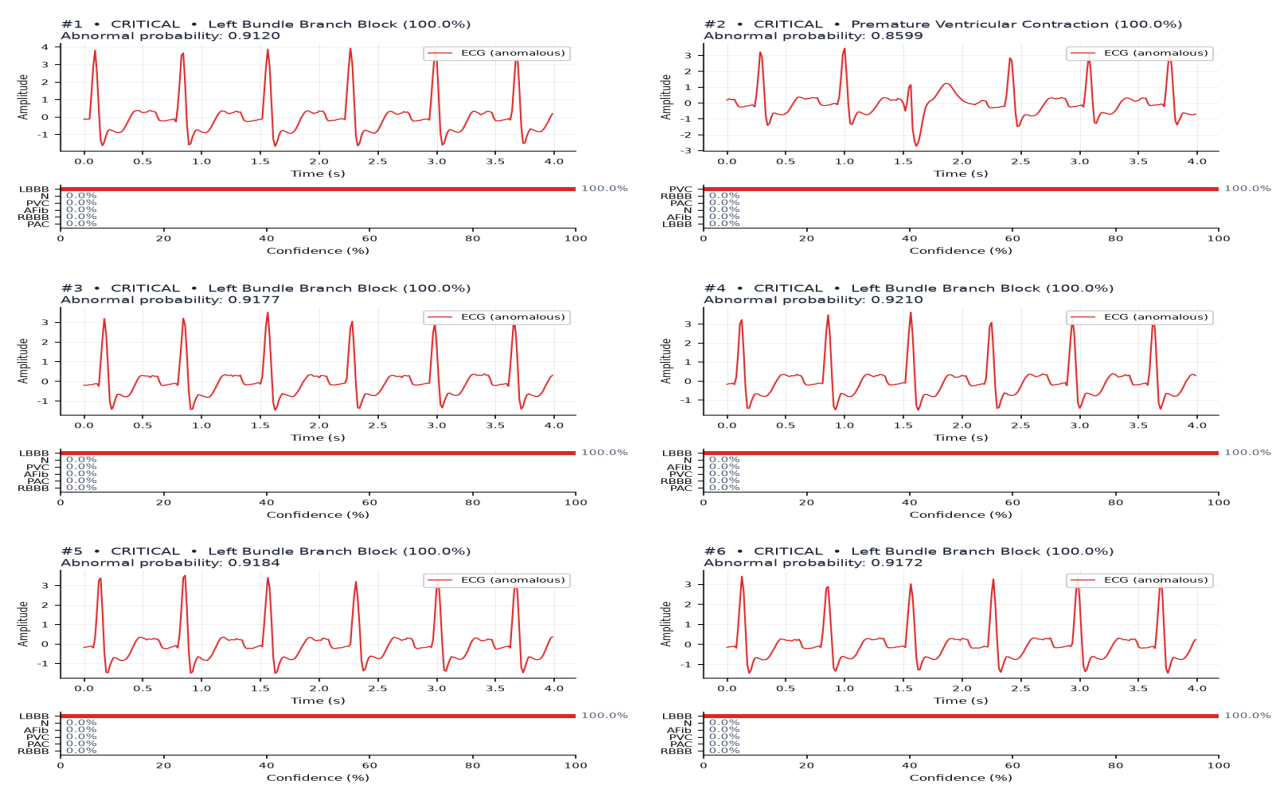
Clinical meaning: Left Bundle Branch Block (LBBB) is a condition where electrical signals to the left ventricle of the heart are delayed. This causes the left ventricle to contract later than the right ventricle, resulting in an abnormal ECG pattern with a wide QRS complex. LBBB affects the timing of the heartbeat but not necessarily the heart's ability to pump blood.

Why it matters: LBBB itself is not life-threatening, but it often indicates underlying heart disease. New-onset LBBB can sometimes signal an acute heart attack and requires immediate medical evaluation. Chronic LBBB (present for a long time) is usually monitored but may require treatment of the underlying condition.

Arrhythmia Type	Detections	Avg Confidence	Example
Left Bundle Branch Block	175	99.9%	score 0.9120, conf 100.0%
Premature Ventricular Contraction	3	100.0%	score 0.8599, conf 100.0%

Detected Anomaly Waveforms

The following plots show R-peak-centred ECG beats that exceeded the supervised abnormal-probability threshold. No reconstruction curve is produced by this feature-based detector. When available, examples are selected from classifications above 50% confidence.



Clinical Summary

100.0% of analysed beats were flagged. This may reflect abnormal morphology, rhythm, noise, or domain shift and requires qualified review. High-confidence classifier findings in this report include: Left Bundle Branch Block, Premature Ventricular Contraction. Only arrhythmia labels with confidence above 50% are included.

Physician Notes

Physician signature: _____

Date: _____

This report is generated automatically by an unsupervised deep-learning anomaly detection system and is intended for decision support only. It does not constitute a medical diagnosis and should not replace clinical judgment by a qualified physician.